

§ 11.7 Curvature and components of acceleration

• $\vec{T}(t) = \frac{\vec{r}'(t)}{\|\vec{r}'(t)\|} = \frac{\vec{v}(t)}{\|\vec{v}(t)\|}$ is the unit tangent vector
(assume $\vec{v}(t) \neq 0$!)

• The curvature is $\kappa = \left\| \frac{d\vec{T}}{dt} \right\| \cdot \frac{1}{\|\vec{v}(t)\|} = \frac{\|\vec{T}'(t)\|}{\|\vec{r}'(t)\|}$

• It depends on the curve, NOT on the parametrization

• Note $\|\vec{v}(t)\| = \frac{ds}{dt}$ where s is the arc length.

$$\text{Thus } \left\| \frac{d\vec{T}}{ds} \right\| = \left\| \frac{d\vec{T}}{dt} \frac{dt}{ds} \right\| = \left\| \frac{d\vec{T}}{dt} \right\| \cdot \frac{1}{\|\vec{v}(t)\|} = \kappa$$

• If $\|\vec{v}(t)\| = 1$ then $\kappa = \|\vec{T}'(t)\|$

Ex 1 $\vec{r}(t) = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix}$ $\vec{r}'(t) = \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix}$ $\vec{T}(t) = \frac{1}{\sqrt{9}} \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix} = \begin{pmatrix} 2/3 \\ -1/3 \\ -2/3 \end{pmatrix}$

$$\kappa = \left\| \frac{d\vec{T}}{dt} \right\| \cdot \frac{1}{\|\vec{v}(t)\|} = \left\| \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\| \cdot \frac{1}{3} = 0$$

Remark For any line $\vec{r}(t) = \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} + t \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ we have $\kappa = 0$

Ex 2 $\vec{r}(t) = (a \cos t, a \sin t)$ $\vec{r}'(t) = (-a \sin t, a \cos t)$
 $\|\vec{v}(t)\| = \sqrt{a^2 \sin^2 t + a^2 \cos^2 t} = a$ (we assume $a > 0$)

$$\vec{T}(t) = \frac{\vec{r}'(t)}{\|\vec{r}'(t)\|} = (-\sin t, \cos t) \quad \vec{T}'(t) = (-\cos t, -\sin t)$$

$$\kappa = \frac{\|\vec{T}'(t)\|}{\|\vec{v}(t)\|} = \frac{1}{a}$$

NB this is the circle $x^2 + y^2 = a^2$

$$\underline{\Sigma x} \quad \vec{r}(t) = (t, a \cos t, a \sin t) \quad \vec{r}'(t) = (1, -a \sin t, a \cos t)$$

$$\|\vec{r}'(t)\| = \sqrt{1+a^2} \quad \vec{T}(t) = \frac{1}{(1+a^2)^{1/2}} (1, -a \sin t, a \cos t)$$

$$\vec{T}'(t) = \frac{1}{(1+a^2)^{1/2}} (0, -a \cos t, -a \sin t)$$

$$K(t) = \frac{\|\vec{T}'(t)\|}{\|\vec{v}(t)\|} = \frac{1}{(1+a^2)} \sqrt{a^2 \cos^2 t + a^2 \sin^2 t} = \frac{a}{1+a^2}$$

$$\underline{\Sigma x} \quad \vec{r}(t) = (2t, t^2)$$

$$\vec{r}'(t) = (2, 2t)$$

$$\|\vec{r}'(t)\| = \sqrt{4+4t^2} = 2(1+t^2)^{1/2}$$

$$\vec{T}(t) = \left(\frac{1}{(1+t^2)^{1/2}}, \frac{t}{(1+t^2)^{1/2}} \right)$$

this is the parabola
 $y = \frac{1}{4} x^2$

$$\vec{T}'(t) = \left(\frac{-1/2 \cdot 2t}{(1+t^2)^{3/2}}, \frac{1}{(1+t^2)^{1/2}} + \frac{t \cdot (-1/2) \cdot 2t}{(1+t^2)^{3/2}} \right) = \left(\frac{-t}{(1+t^2)^{3/2}}, \frac{1+t^2-t^2}{(1+t^2)^{3/2}} \right)$$

$$K(t) = \frac{\|\vec{T}'(t)\|}{\|\vec{r}'(t)\|} = \frac{1}{2(1+t^2)^{1/2}} \cdot \frac{1}{(1+t^2)^{3/2}} \cdot \sqrt{t^2+1} = \frac{1}{2(1+t^2)^{3/2}}$$

MAXIMUM of $K(t)$ is $K(0) = \frac{1}{2}$

$$\underline{\Sigma x} \quad \vec{r}(t) = (x(t), y(t)) \quad \vec{r}'(t) = (x', y') \quad \|\vec{r}'\| = \sqrt{(x')^2 + (y')^2}$$

$$\vec{T}(t) = \left(\frac{x'}{\sqrt{(x')^2 + (y')^2}}, \frac{y'}{\sqrt{(x')^2 + (y')^2}} \right)$$

$$\vec{T}'(t) = \left(\frac{x''}{\sqrt{(x')^2 + (y')^2}} - \frac{1}{2} \frac{(2x'x'' + 2y'y'')x'}{((x')^2 + (y')^2)^{3/2}}, \frac{y''}{\sqrt{(x')^2 + (y')^2}} - \frac{1}{2} \frac{(2x'x'' + 2y'y'')y'}{((x')^2 + (y')^2)^{3/2}} \right)$$

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... Many calculations later

$$K = \frac{|x'y'' - y'x''|}{((x')^2 + (y')^2)^{3/2}}$$

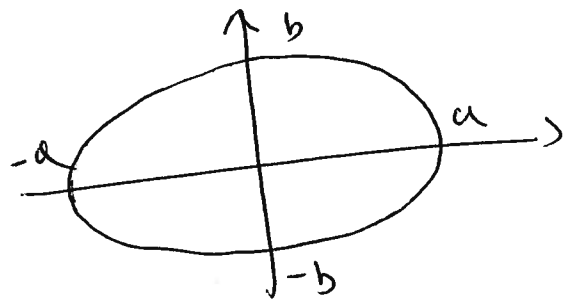
Ex $x = a \cos t$ $y = b \sin t$

assume $a \geq b > 0$

$$x' = -a \sin t \quad y' = b \cos t$$

$$x'' = -a \cos t \quad y'' = -b \sin t$$

$$\text{so } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$



$$K(t) = \frac{|+ab \sin^2 t + ba \cos^2 t|}{(a^2 \sin^2 t + b^2 \cos^2 t)^{3/2}}$$

$$= \frac{ab}{(a^2 - b^2) \sin^2 t + b^2}^{3/2}$$

MAX at $t = 0, \pi$

$$K(0) = K(\pi) = \frac{ab}{b^3} = \frac{a}{b^2}$$

MIN at $t = \pi/2, 3\pi/2$

$$K(\pi/2) = \frac{ab}{a^3} = \frac{b}{a^2}$$

Note that $\vec{T}(t) \cdot \vec{T}(t) = 1$ so

$$2\vec{T}(t) \cdot \vec{T}'(t) = \vec{T}'(t) \cdot \vec{T}(t) + \vec{T}(t) \cdot \vec{T}'(t) = 0$$

so $\vec{T}(t) \cdot \vec{T}'(t) = 0$ i.e. $\vec{T}(t)$ is perpendicular to $\vec{T}'(t)$

Define $\vec{N}(t) = \frac{\vec{T}'(t)}{\|\vec{T}'(t)\|}$ the unit normal vector

It turns out that we can write

$$\vec{a} = a_T \vec{T} + a_N \vec{N}$$

ie. $\vec{a}$ has 2 components: a tangential one & a normal one

Recall $\vec{v} = \frac{ds}{dt} \cdot \vec{T}$ (or $\vec{T} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{\vec{v}}{ds/dt}$)

$$\text{so } \vec{v}' = \frac{ds}{dt} \vec{T}' + \frac{d^2s}{dt^2} \vec{T} = \frac{ds}{dt} \|\vec{T}'(t)\| \cdot \vec{N}(t) + \frac{d^2s}{dt^2} \vec{T}(t)$$

$$= \left(\frac{ds}{dt}\right)^2 \kappa \vec{N} + \frac{d^2s}{dt^2} \vec{T}$$

$$= \underset{\substack{\uparrow \\ \text{normal} \\ \text{acceleration}}}{a_N} \cdot \vec{N} + \underset{\substack{\uparrow \\ \text{tangential} \\ \text{acceleration}}}{a_T} \cdot \vec{T}$$

We use $\kappa = \frac{\|\vec{T}'(t)\|}{ds/dt}$

so $\|\vec{T}'(t)\| = \kappa \frac{ds}{dt}$

$$\|\vec{a}\|^2 = (a_N \vec{N} + a_T \vec{T}) \cdot (a_N \vec{N} + a_T \vec{T}) = a_N^2 \vec{N} \cdot \vec{N} + 2a_N a_T \vec{N} \cdot \vec{T} + a_T^2 \vec{T} \cdot \vec{T} \\ = a_N^2 + a_T^2$$

$$\text{so } a_N^2 = \|\vec{a}\|^2 - a_T^2$$

$\uparrow \quad \nearrow$
easier to compute

$$a_N = \left(\frac{ds}{dt}\right)^2 \kappa = \|\vec{r}'\|^2 \cdot \kappa \quad a_T = \frac{d^2s}{dt^2} = \frac{d}{dt} \|\vec{r}'\|$$

We have $\vec{a} \cdot \vec{T} = (a_T \vec{T} + a_N \vec{N}) \cdot \vec{T} = a_T$ (as $\vec{T} \cdot \vec{T} = 1$, $\vec{N} \cdot \vec{T} = 0$)

so
$$a_T = \vec{a} \cdot \vec{T} = \frac{\vec{r}'' \cdot \vec{r}'}{\|\vec{r}'\|} = \frac{\vec{v} \cdot \vec{v}'}{\|\vec{v}'\|}$$

Also $\vec{T} \times \vec{a} = \vec{T} \times (a_T \vec{T} + a_N \vec{N}) = a_N \vec{T} \times \vec{N}$

$\|\vec{T} \times \vec{a}\| = a_N \cdot \|\vec{T} \times \vec{N}\| = a_N \cdot \|\vec{T}\| \cdot \|\vec{N}\| = a_N$

so
$$a_N = \|\vec{T} \times \vec{a}\| = \left\| \frac{\vec{r}'}{\|\vec{r}'\|} \times \vec{r}'' \right\| = \frac{\|\vec{r}' \times \vec{r}''\|}{\|\vec{r}'\|}$$

$$\kappa = \frac{a_N}{(ds/dt)^2} = \frac{\|\vec{r}' \times \vec{r}''\|}{\|\vec{r}'\| \|\vec{r}'\|^2} = \frac{\|\vec{r}' \times \vec{r}''\|}{\|\vec{r}'\|^3}$$

The vector $\vec{B} = \vec{T} \times \vec{N}$ is the binormal vector

Ex $\vec{r}(t) = (t, t^2, \frac{t^3}{3})$ Find $\vec{T}, \vec{N}, \vec{B}, a_T, a_N, \kappa$, at $t=1$

$\vec{r}'(t) = (1, 2t, t^2)$ $\vec{r}''(t) = (0, 2, 2t)$ $\|\vec{r}'(t)\| = \sqrt{1+4t^2+t^4}$

$\vec{T}(1) = \frac{1}{\sqrt{6}} (1, 2, 1)$ $\vec{r}'(1) = (1, 2, 1)$ $\vec{r}''(1) = (0, 2, 2)$

$a_T = \frac{\vec{r}' \cdot \vec{r}''}{\|\vec{r}'\|} = \frac{1 \cdot 0 + 2 \cdot 2 + 1 \cdot 2}{\sqrt{6}} = \frac{6}{\sqrt{6}} = \sqrt{6}$

$a_N = \frac{\|\vec{r}' \times \vec{r}''\|}{\|\vec{r}'\|} = \frac{1}{\sqrt{6}} \left\| \begin{vmatrix} i & j & k \\ 1 & 2 & 1 \\ 0 & 2 & 2 \end{vmatrix} \right\| = \frac{1}{\sqrt{6}} \|(2, -2, 2)\| = \frac{2\sqrt{3}}{\sqrt{6}} = \sqrt{2}$

$\vec{a} = a_N \vec{N} + a_T \vec{T} \Rightarrow \vec{N} = \frac{\vec{a} - a_T \vec{T}}{a_N} = \frac{(0, 2, 2) - \frac{\sqrt{6}}{\sqrt{6}} (1, 2, 1)}{\sqrt{2}} = \frac{(-1, 0, 1)}{\sqrt{2}}$

$\vec{B} = \vec{T} \times \vec{N} = \begin{vmatrix} i & j & k \\ \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{vmatrix} = \left(\frac{2}{\sqrt{12}}, \frac{-2}{\sqrt{12}}, \frac{2}{\sqrt{12}} \right)$ $\kappa = \frac{\|\vec{r}' \times \vec{r}''\|}{\|\vec{r}'\|^3} = \frac{a_N}{\|\vec{r}'\|^2} = \frac{\sqrt{2}}{6} = \frac{1}{3\sqrt{2}}$

Exercises for 11.7

Ex Find $\kappa, T, N, B, a_N, a_T$ for

- $r(t) = (3t, 3t^2, 1)$ at $t = 1/3$
- $r(t) = (e^{-t}, 2t, e^t)$ at $t = 0$
- $r(t) = (\sin 3t, \cos 3t, 2t)$ at $t = \pi/3$
- $r(t) = (e^{-2t}, e^{2t}, 2\sqrt{2}t)$ at $t = 0$
- $r(t) = (t, \frac{1}{3}t^3, t^{-1})$ at $t = 1$

Ex Show that $\frac{d\vec{B}}{ds}$ is perpendicular to $\vec{T}$.